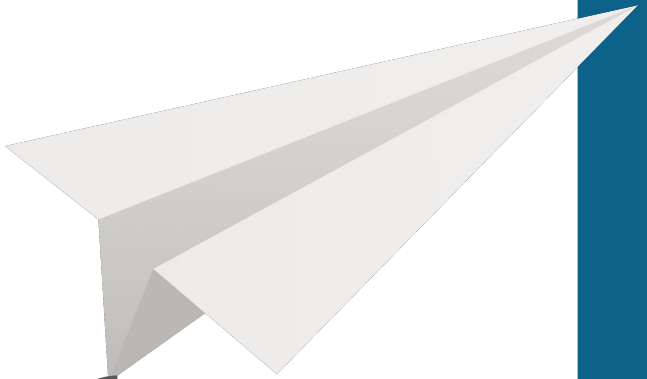
WEB ANALYTICS · GROUP 10

FLEXIBLE FLIGHT FINDER



INDEX:

1. PROJECT OVERVIEW
2. STATE-OF-THE-ART
3. DATA ACQUISITION PIPELINE
4. METHODOLOGY & DATA ANALYSIS STRATEGY
5. DASHBOARD & VISUALIZATION
6. CONCLUSIONS

PROJECT OVERVIEW



PROBLEM:

- Traditional flight search engines require users to specify exact dates and fixed destinations.

- This becomes a limitation for travelers with flexible schedules or those simply looking for the cheapest time or place to fly.



"If I am flexible with my dates and destination, when or where should I fly to get the lowest price?"



GOAL: From FLEXIBLE search to the BEST DEAL

Transform flexible search criteria into **Actionable Insights**

INPUT (CRITERIA) → PROCESS (DATA ANALYSIS) → OUTPUT (RESULT)

Build an analytical tool (**dashboard**) that scrapes flight data and provides users with a dynamic, visual overview of the cheapest destinations and best travel dates within a defined flexible search space.



STATE-OF-THE-ART



Many search engines already address **flexibility:**

	EXISTING TOOLS (Skyscanner,...)	OUR UNIQUE VALUE
Google Flights Skyscanner Amadeus	It shows the lowest daily or monthly price Users need to know where and when to fly Low Cost Airlines are not taken into account	Provides a deep analysis of the flight structure network Not knowing where or when to fly can let us discover non-obvious patterns We use this low-cost airlines to indeed find cheap flights



DATA ACQUISITION





INITIAL DATA SOURCES

MAIN DATA SOURCE?



AMADEUS API

ACCESIBLE?



SKYSCANNER API



GOOGLE FLIGHTS

FINAL DATA SOURCES



GOOGLE FLIGHTS

KAYAK



AMADEUS

FINAL DATA SOURCES

SCRAPING

over 400.000 flights

GOOGLE FLIGHTS

KAYAK

API

flight inspiration

AMADEUS



KAYAK SCRAPING





SCRAPING

- For each flight card:
  - Extracted airline
  - Departure and arrival times
  - Number of stops
  - Price
  - Duration

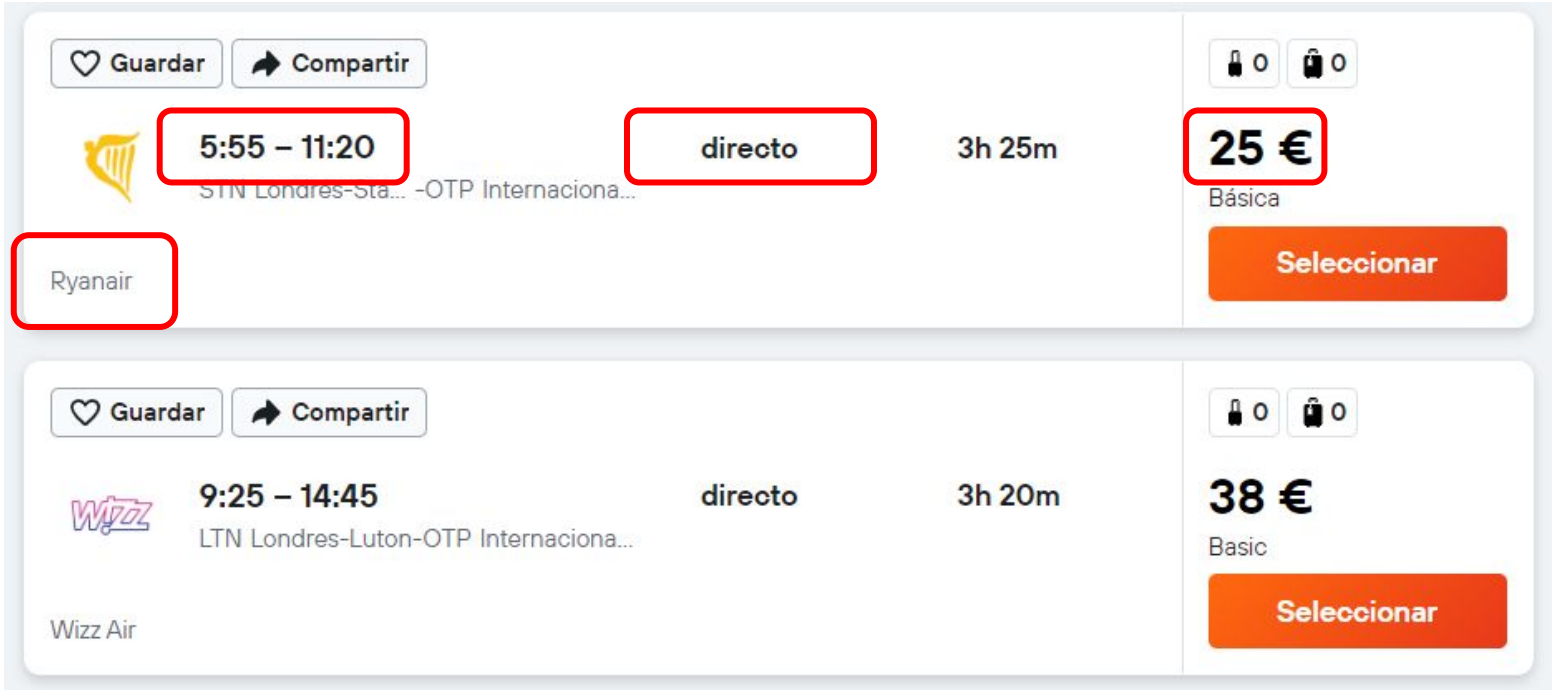
DATA STORAGE

- Collected all flight entries into a Python list.
- Converted to Pandas DataFrame.
- Exported the final dataset to CSV

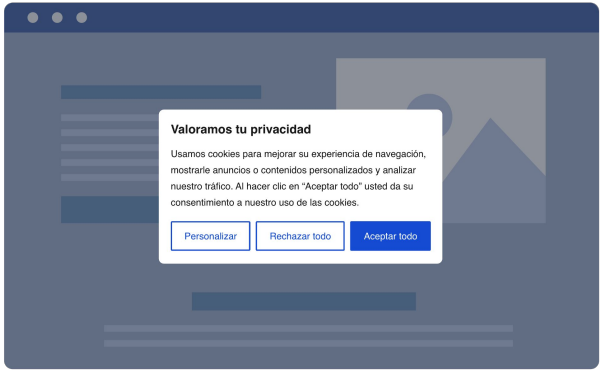


ANTIBLOCKING CONSIDERATIONS

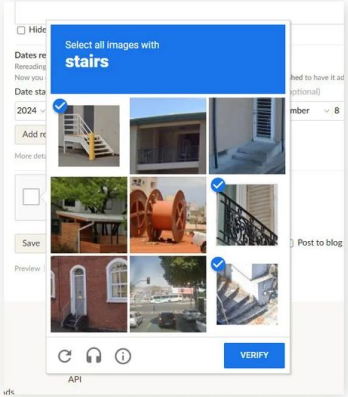
- Introduced random sleeps between requests.
- Avoided headless mode.
- Used scrolling to ensure all flight cards loaded.


Guardar
Compartir
0
0
5:55 – 11:20
STN Londres-Sta... -OTP Internaciona...
directo
3h 25m
25 €
Básica
Ryanair
Seleccionar
Guardar
Compartir
0
0
9:25 – 14:45
LTN Londres-Luton-OTP Internaciona...
directo
3h 20m
38 €
Basic
Wizz Air
Seleccionar

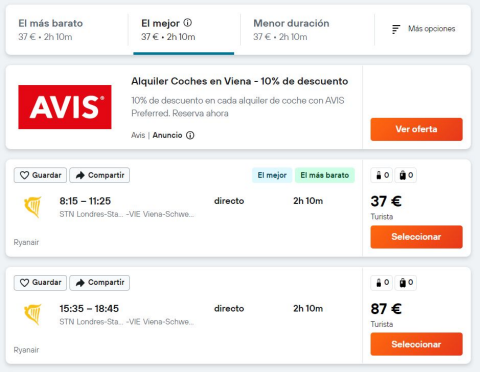
CHALLENGES



HANDLING WITH POP-UPS



BOT DETECTION



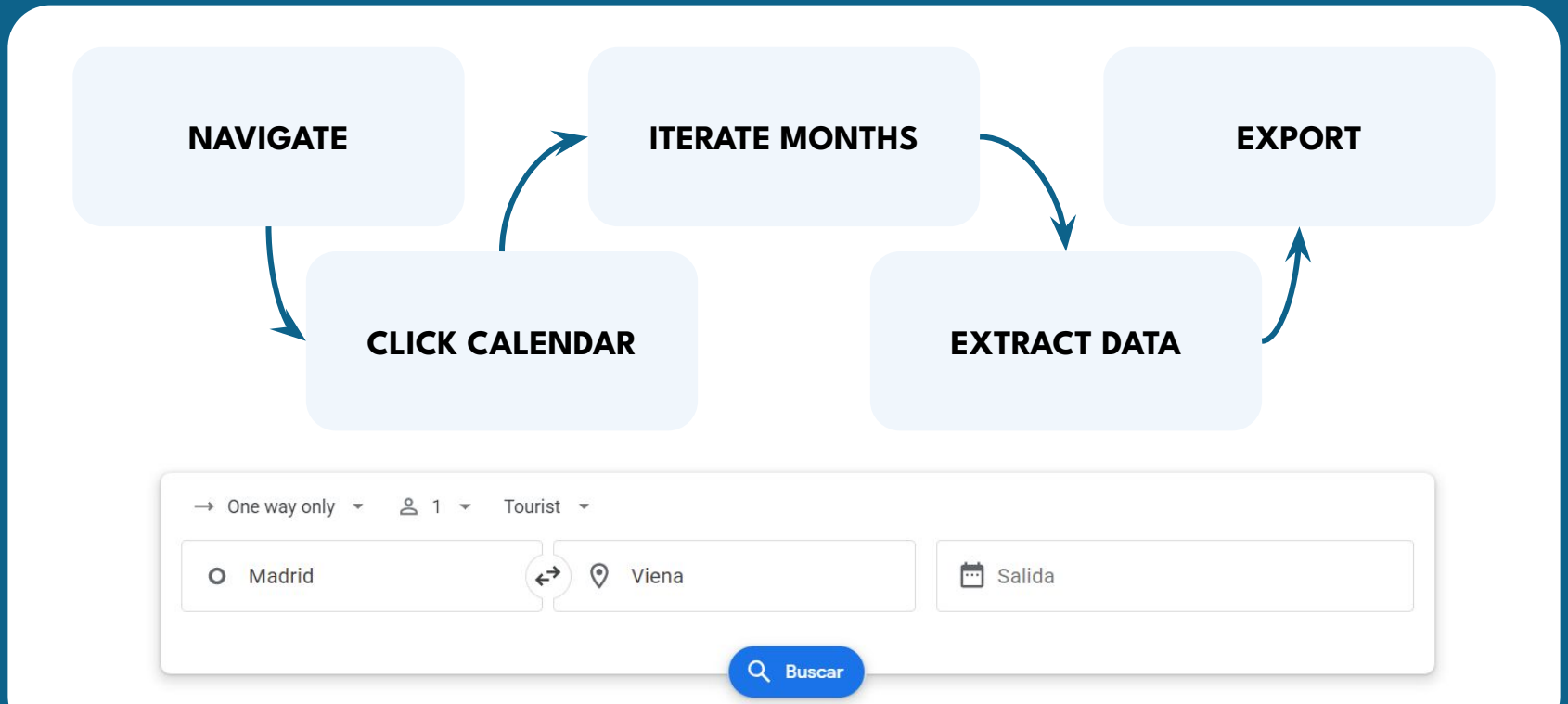
FILTERING ADS

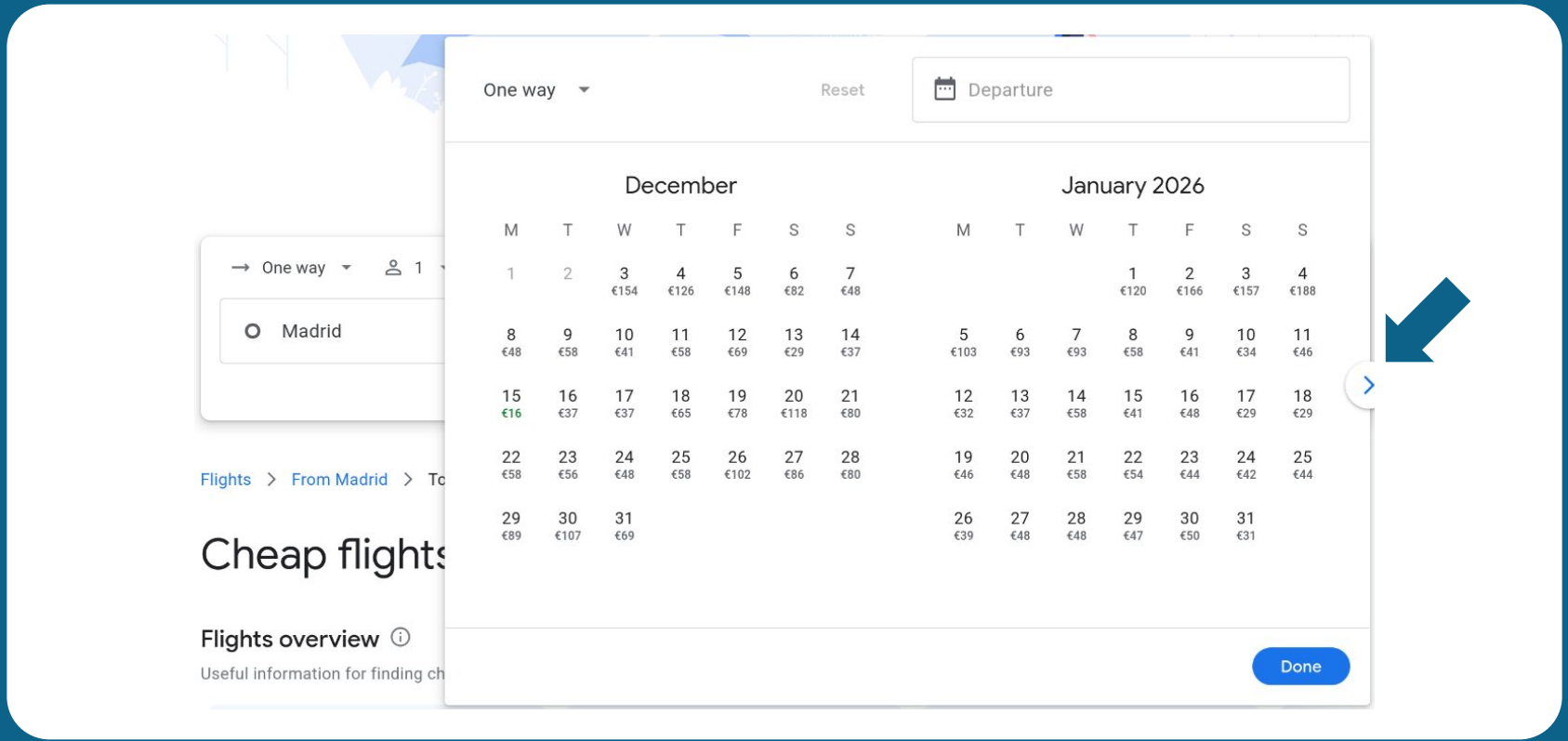


GOOGLE FLIGHTS SCRAPING






NAVIGATE
ITERATE MONTHS
EXPORT
CLICK CALENDAR
EXTRACT DATA
One way only
1
Tourist
Madrid
Viena
Salida
Buscar


One way
Reset
Departure
December
January 2026
M T W T F S S
1 2 3 €154 4 €126 5 €148 6 €82 7 €48
8 €48 9 €58 10 €41 11 €58 12 €69 13 €29 14 €37
15 €16 16 €37 17 €37 18 €65 19 €78 20 €118 21 €80
22 €58 23 €56 24 €48 25 €58 26 €102 27 €86 28 €80
29 €89 30 €107 31 €69
1 €120 2 €166 3 €157 4 €188
5 €103 6 €93 7 €93 8 €58 9 €41 10 €34 11 €46
12 €32 13 €37 14 €58 15 €41 16 €48 17 €29 18 €29
19 €46 20 €48 21 €58 22 €54 23 €44 24 €42 25 €44
26 €39 27 €48 28 €48 29 €47 30 €50 31 €31
Done
One way
1
Madrid
Flights
From Madrid
Cheap flights
Flights overview
Useful information for finding ch



AMADEUS SEARCH API





Really good source since has real-time price values of flights offering features such as price, origin, destination...

LIMITATIONS



- Limited free calls (2000 per month) for the **test** environment
- Low-cost airlines are not included
- “Pay-per-use” approach: high cost

Flight Inspiration Search

- What are the cheapest cities to fly to from *Madrid?*
- used to implement our own Flight Inspiration Search





FLIGHT PRICE PREDICTION MODEL





CONSIDERATIONS



almost **500,000 flights** as data

random **70/30 training/test split**

XGBoost model in Python

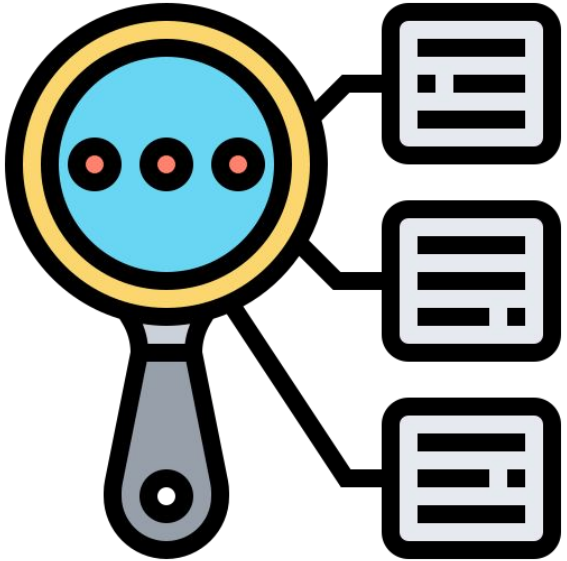
average error of **12.97€** on test set

- prediction for untrained routes significantly worse
- unknown routes test sets: average errors of up to **30€**



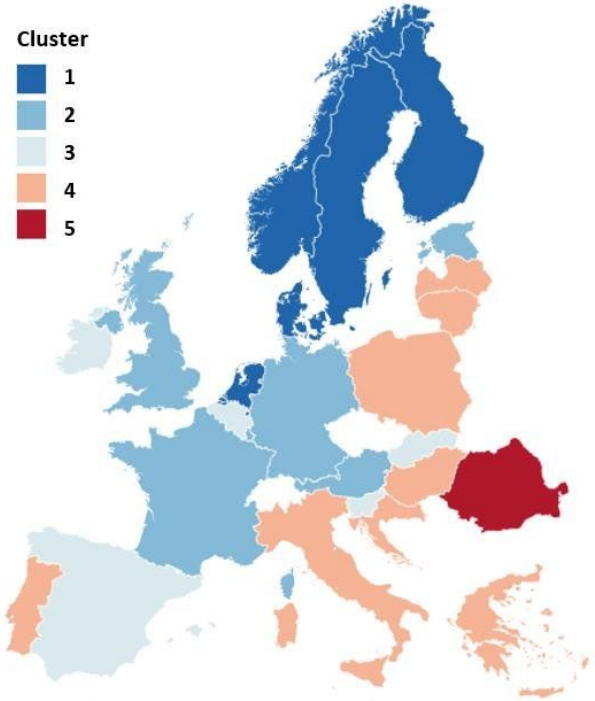
DATA CLASSIFICATION





We have built a complete **clustering** pipeline to analyze how European cities relate to each other based on real flight data. This analysis includes:

- ★ Geographical distance
- ★ Cheap-flight connectivity
- ★ Graph structure & network centrality
- ★ Node2Vec structural embeddings
- ★ Tabular flight features
- ★ PCA dimensionality reduction
- ★ Hybrid KMeans clustering



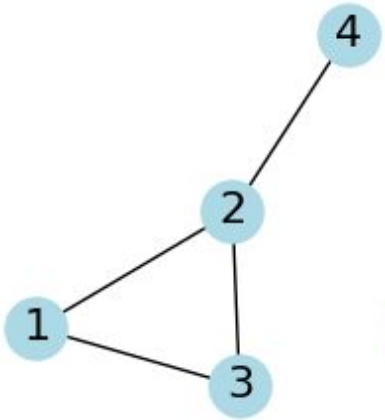
DISTANCE-BASED CLUSTERING

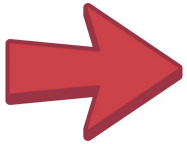
“How would cities group if we only looked at geography?”

AGGLOMERATIVE CLUSTERING

SPECTRAL CLUSTERING

MODELLING EUROPE AS A GRAPH







NODE

EDGE

“How do low-cost routes partition Europe into travel zones, and which cities act as key connectors?”

LOUVAIN COMMUNITY DETECTION

- identify communities: groups of cities that form dense cheap-flight regions.

These communities reveal travel zones created by airline economics rather than physical distance.

BETWEENNESS CENTRALITY

- identify strategic hubs that act as bridges between regions.

These are cities with high structural importance for multi-city routing and travel efficiency.

DEEPER PATTERNS WITH NODE2VEC

“Which cities behave similarly in the graph, even if they are geographically distant?”

Node2Vec Embeddings (Hidden Structural Features)

Neighborhood structure	**Cheap-route similarity**	**Community membership**	**Connectivity importance**

Having computed some metrics as minimum price, price-per-km... and degree, closeness...

Hybrid Feature Construction + PCA

“What deeper patterns emerge when combining all travel, price, and structural data?”

Cluster 0 — High-cost Peripheral Outliers

Cities such as Lisbon, Stockholm, Zurich
→ expensive, low-connectivity, geographically peripheral

Cluster 1 — Balanced Mid-range Connectors

Cities such as Amsterdam, Vienna, Madrid, Budapest
→ stable, moderately priced, well-connected but not dominant

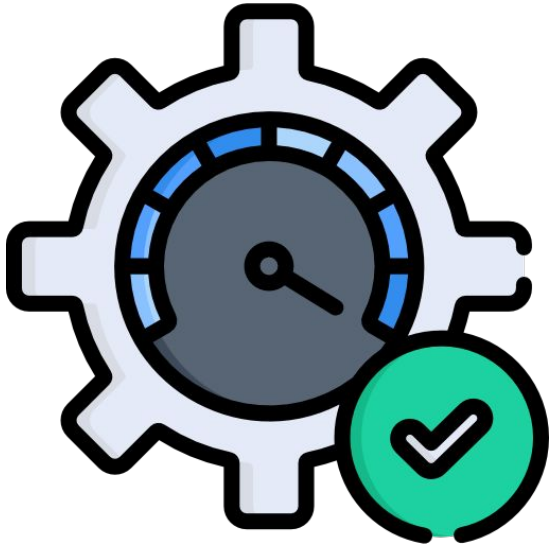
Cluster 2 — Major European Hubs

Cities like London, Paris, Berlin, Rome, Istanbul
→ highly connected, structurally central, competitively priced



OPTIMIZER MODEL





AUTOMATIC MULTI-CITY TRAVEL PLAN

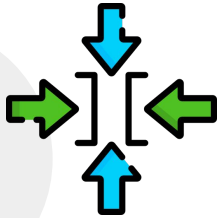
SELECTS WHICH CITIES TO VISIT

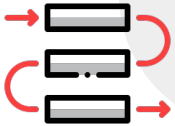


SELECTS VALID FLIGHTS BETWEEN CITIES



RESPECTS DATES, BUDGET AND MINIMUM STAY CONSTRAINTS





DECIDES THE BEST SEQUENCE TO VISIT



MAXIMIZES NUMBER OF VISITED CITIES



GUARANTEES A SINGLE ROUTE AND RETURN TO ORIGIN CITY

AUTOMATIC MULTI-CITY TRAVEL PLAN



SELECTS WHICH CITIES TO VISIT



SELECTS VALID FLIGHTS BETWEEN



RESPECTS DATES, BUDGET AND MINIMUM STAY

LINEAR INTEGER PROGRAMMING



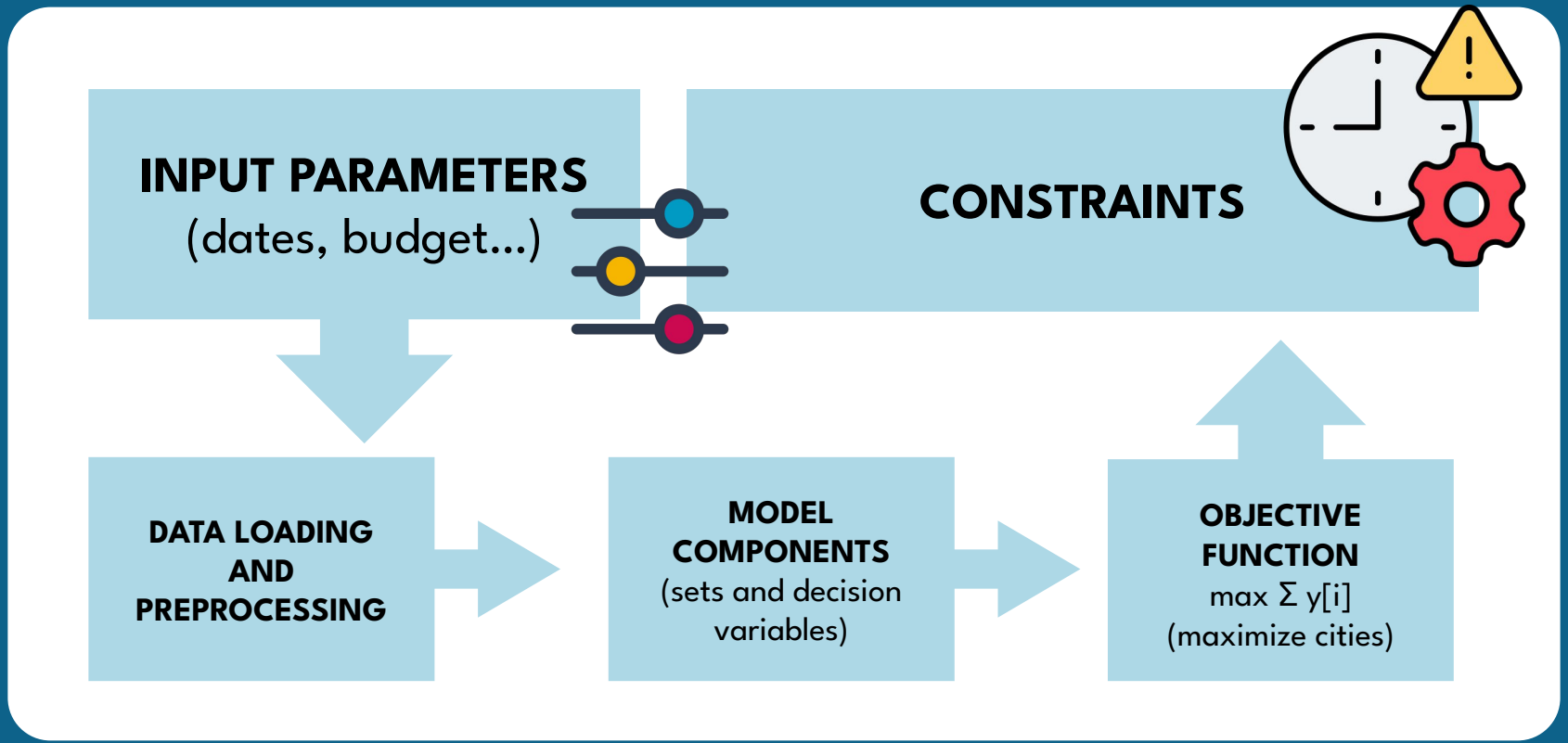
DECIDES THE BEST SEQUENCE TO VISIT



MAXIMIZES NUMBER OF VISITED CITIES



GUARANTEES A SINGLE ROUTE AND RETURN TO ORIGIN CITY

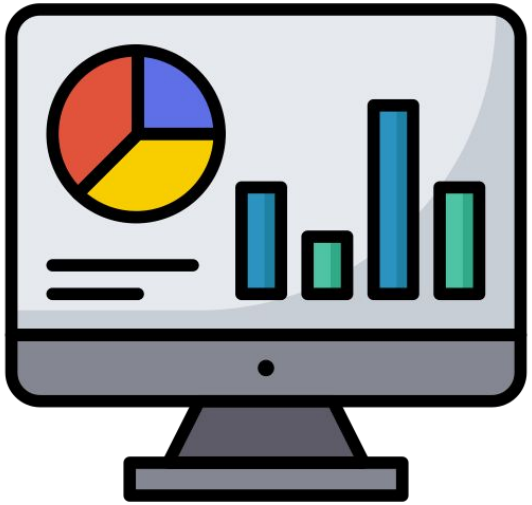

INPUT PARAMETERS
(dates, budget...)
CONSTRAINTS
DATA LOADING AND PREPROCESSING
MODEL COMPONENTS
(sets and decision variables)
OBJECTIVE FUNCTION
max Σ y[i]
(maximize cities)

SOME OF THE CONSTRAINTS...

1. Flow Constraints:
    - If a city is visited, exactly one incoming and one outgoing flight must exist.
2. Origin Constraints:
    - Exactly one departure from the origin.
    - Exactly one return to the origin.
3. Budget Constraint:
4. Minimum Days per City
5. Total Days Constraint
6. MTZ Subtour Elimination:
7. Month Window Constraints:



DASHBOARD



Home

Flexible Flight Finder

Discover the best flight deals across Europe with interactive maps and comprehensive statistics

400,000+ flights analyzed

Find Your Next Adventure at the Best Price

Explore interactive visualizations, compare prices across routes, and discover the cheapest times to fly to your dream destination.

Flight Inspiration Search

Flexible search for the best deals

Origin City: Madrid

Date Range: 30. 12. 2025 to 15. 03. 2026

Trip Duration (Days): 4 to 7

Maximum Price (€): 400

Non-stop only | One way | Multi-city trip

Search Flights

Interactive Map Explorer

Visualize flight routes on an interactive map

- Explore routes from any city
- Compare prices across destinations

Start Exploring → Maps Routes

Statistics Dashboard

Analyze pricing trends and discover the best deals

- Calendar heatmap of prices
- Top cheapest routes ranked

View Statistics → Charts Analytics

City Classification

Explore clustering patterns across European cities

- Network community detection
- ML-based hybrid clustering

Explore Patterns → ML Clustering

Powered by scraped flight data • European routes only

Home > Flight Results

Flight Results

From Madrid | 2025-12-30 - 2026-03-15 | 4-7 days | Max €400 | Multi-city

Optimized Multi-City Trip

4 cities • 4 flights

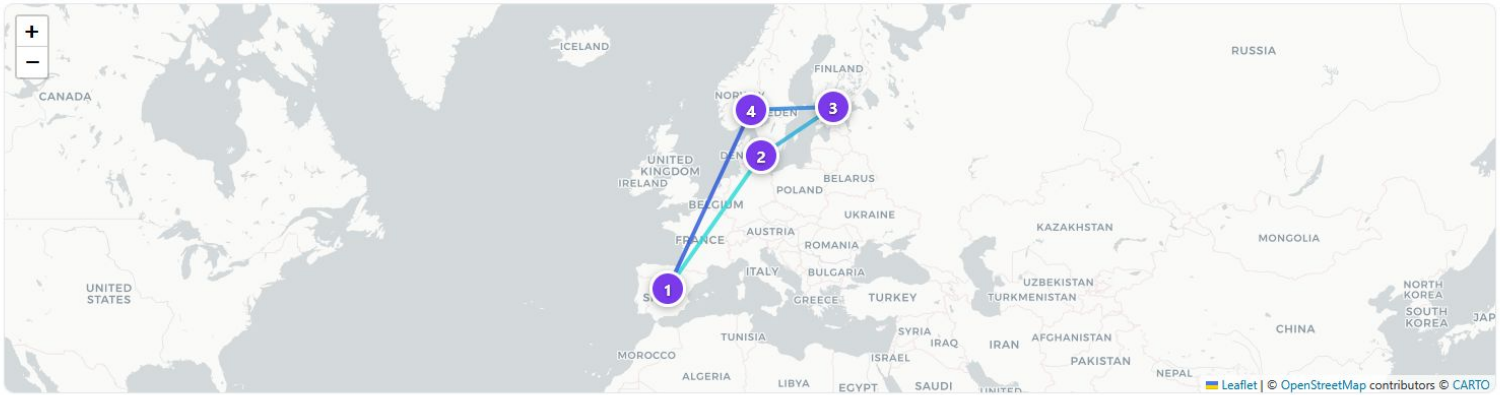
Copenhagen → Helsinki → Madrid → Oslo

€396

Total trip cost

Best Route

Route Map



Map Explorer

Flexible Flight Finder

Click on cities to explore routes • Click on routes to view cheapest flights



Madrid → Warsaw

Top 10 cheapest flights (267 total)

- €25 — 2293 km — 2026-02-13 (Friday)
- €34 — 2293 km — 2026-02-06 (Friday)
- €35 — 2293 km — 2026-03-13 (Friday)
- €36 — 2293 km — 2026-03-06 (Friday)
- €39 — 2293 km — 2026-02-08 (Sunday)
- €40 — 2293 km — 2026-02-03 (Tuesday)
- €40 — 2293 km — 2026-02-04 (Wednesday)
- €40 — 2293 km — 2026-02-05 (Thursday)
- €40 — 2293 km — 2026-02-07 (Saturday)

Statistics Dashboard

Flight Statistics Dashboard

Discover the cheapest flight deals and pricing trends

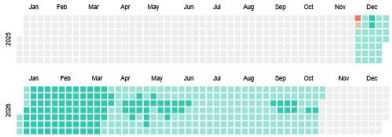
Start Date: dd. mm. yyyy

End Date: dd. mm. yyyy

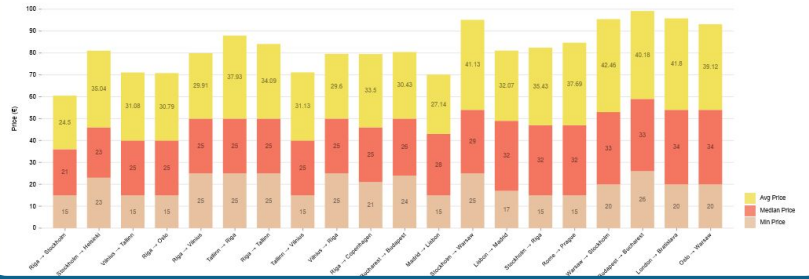
Refresh | Clear Filters

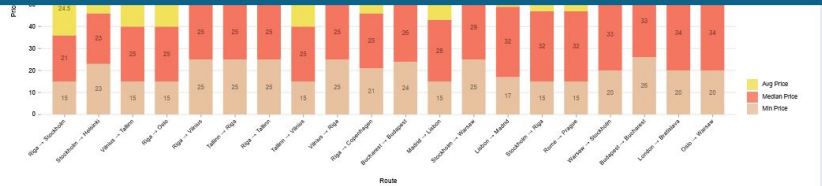
160 208 flights | 600 routes | 2025-11-30 to 2026-10-26 | ✓ Cached data

Daily Median Prices - Calendar Heatmap

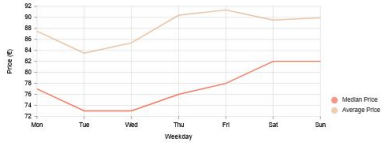


Top 20 Cheapest Routes by Median Price

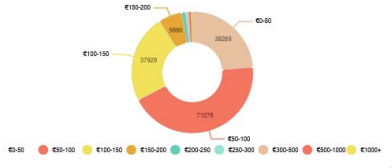




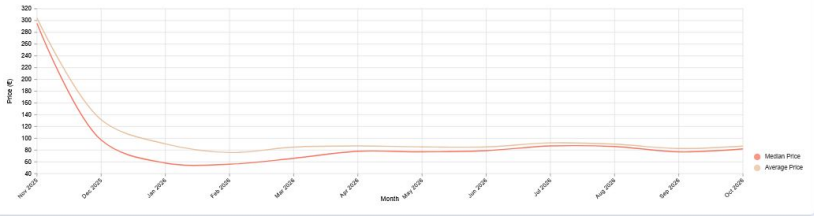
Price Trends by Weekday



Price Distribution



Monthly Price Trends



Home > Flight Results

Flight Results

From Madrid | 2025-12-30 - 2026-03-15 | 4-7 days | Max €400

Madrid → Porto

Departure: Wed, Jan 7
Return: Wed, Jan 14
Duration: 7 days
Distance: 438 km

€52
per person
Price expected to be stable in the next 7 days
Select Flight

Madrid → Lisbon

Departure: Sun, Jan 4
Return: Sun, Jan 11
Duration: 7 days
Distance: 513 km

€54
per person
Price expected to rise in the next 7 days
Select Flight

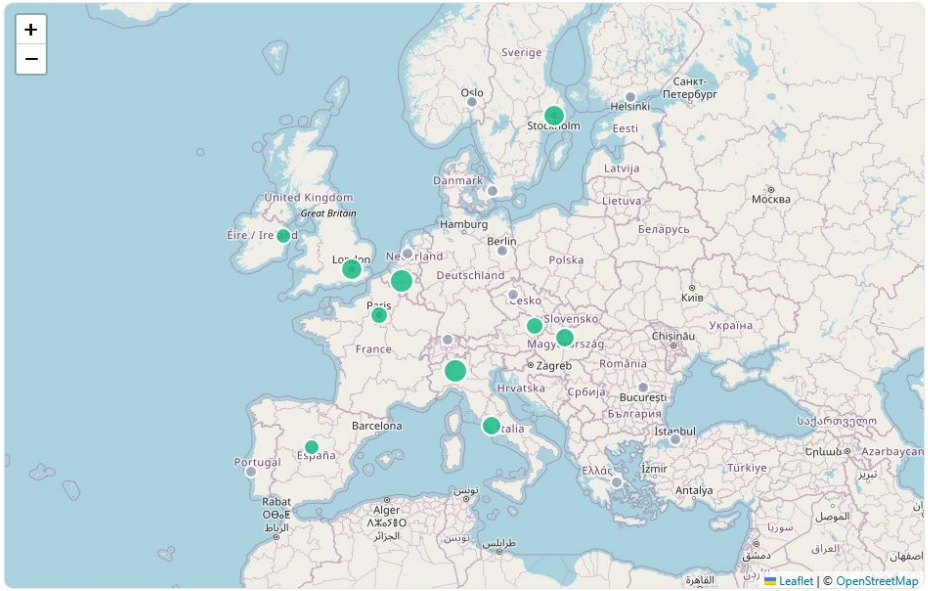
Madrid → Palma De Mallorca

€56
per person
Price expected to fall in the next 7 days

Select Classification Method

Geographical	Network Communities	Budget Hubs	Hybrid Clustering
Distance-based regions	Cheap-flight connections	Strategic connectors	ML-based patterns

Interactive Map



Legend

- Budget Hub
- Other City

Size indicates importance ranking

About This Classification

Betweenness Centrality

Cities ranked by how often they serve as bridges in the cheap-flight network. Higher betweenness indicates more strategic importance.

Top Budget Hubs:

1. Milan (Most central)
2. Brussels
3. Stockholm
4. London
5. Rome

These cities are optimal for multi-city routing and budget travel planning.

CONCLUSIONS



We succeeded in answering our initial question:

"If I am flexible with my dates and destination, when or where should I fly to get the lowest price?"

We achieved the flexible finding (Origin → Any Destination, Date → Any time period) and comparison of prices

We were able to create a tool which helps us to compare prices, find for flexible flights (destination, time...), see the main insights and when prices increase/decrease

We used forecasting to see if the price is intended to increase/decrease in the following days

We could create trips with a constraint on time and budget

Better predictions need more data, and even then can be unpredictable



- data collection can be expensive and painful
- start collecting data early (even before knowing what to do with data) and as much as possible
- results only “visible” in presentation of data



- Persistent data pipeline to ensure continuous data collection over several months



THANK YOU FOR YOUR ATTENTION

ANY QUESTIONS?